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Accurate classification of *BRCA1* variants with saturation genome editing

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SUPPLEMENTARY INFORMATION

SUPPLEMENTARY NOTES

Supplementary Note 1

We leveraged multiple lines of evidence along with SGE sequence-function maps in reviewing the three SNVs for which our classifications were clearly discordant with ClinVar. Discordant SNVs that we assayed in our preliminary experiments in WT HAP1 cells had similar scores, suggesting their SGE-based classifications are unlikely due to noise in our assay or the use of *LIG4*-deficient cells (**Extended Data Fig. 10c**).

One missense SNV designated ‘pathogenic’ in ClinVar that we scored as functional, c.5359T>A (C1787S) (score = 0.069), was identified through segregation with disease. However, in each case, it was seen in *cis* with a second SNV at the neighboring amino acid position⁵¹. Our data as well as data from other functional assays¹² suggest c.5359T>A on its own is functional. The linked SNV c.5363G>A (G1788D) also scored as functional (-0.693), but very near to the intermediate function category (-0.748). It is possible that the two variants together are damaging, but our results suggest if c.5359T>A (C1787S) is encountered independently it should not be considered pathogenic.

A second disagreement was identified in the exon 2 splice acceptor, c.-19-2A>G. This SNV was annotated as ‘pathogenic’ in ClinVar based on its occurrence at a splice acceptor site⁵², rather than from having been associated with disease. None of the 6 SNVs that we introduced here scored as non-functional (5 functional, 1 intermediate). Exon 2 contains the *BRCA1* translation initiation codon, meaning that alternate splice forms may preserve the complete open reading frame. Of note, CADD scores for SNVs across the exon 2 acceptor site were much lower than for SNVs in other canonical splice sites (**Extended Data Fig. 10d**), consequent to a lack of sequence conservation (**Extended Data Fig. 10e**). Further supporting that this splice site is not essential for *BRCA1* function, RNA sequencing from both blood and breast tissue reveals substantial variability at this junction, including an alternative acceptor site upstream of the initiation codon^{49,50}. In the targeted RNA-seq data from exon 2 SGE experiments, this alternative site is used in ~4% of transcripts spliced from the canonical exon 1 donor site (although SGE cannot discern if acceptor site variants contribute disproportionately to this spliceform). While we cannot rule out modest reductions in mRNA caused by these SNVs, this evidence collectively suggests that this acceptor site is dispensable in our assay, and variants in it should not be assumed pathogenic without additional evidence.

Exon 16 harbored the third discordantly classified SNV, the ‘benign’ c.5044G>A (E1682K) variant, which scored as non-functional in our assay. Of note, c.5044G>A resides in a predicted exonic splice enhancer (ESE)³⁷, and its low function score was substantiated by a greater than 10-fold reduction in RNA (**Extended Data Fig. 10a**). Neighboring SNVs in the predicted ESE also reduced RNA expression, corroborating the element’s importance. However, previous assays of this variant did not measure splicing consequences¹². Although this missense SNV is rare (absent from gnomAD and Bravo), reports indicate it was designated as benign in part because it was deemed to exist in *trans* with a variant considered pathogenic⁵³. Biallelic *BRCA1* loss-of-function mutations are thought to be embryonic lethal, although notable exceptions have

been documented^{54,55}. The molecular and phenotypic data required to more fully investigate this discrepancy are not publicly available¹².

Supplementary Note 2

There was a modest but significant difference between RNA scores of functional missense SNVs and those of non-functional missense SNVs. Only considering missense SNVs with RNA scores greater than -1 (*i.e.* those unlikely to reduce function through splicing defects), the median RNA score among 1,388 functional missense SNVs was -0.01, while the median RNA score of the 346 non-functional missense SNVs was -0.365 (Wilcoxon Rank Sum Test $P = 1.2 \times 10^{-54}$). This finding is consistent with missense SNVs that cause loss of BRCA1 function producing less BRCA1 mRNA as mitotic activity is reduced.

Supplementary Note 3

Our study has several caveats. First, the exact requirements for *BRCA1* function essential to maintaining *in vitro* viability and growth of HAP1 cells, as opposed to mediating *in vivo* tumor suppression, are not known. For instance, we cannot rule out differences in splicing or dosage requirements between our *in vitro* model vs. *in vivo* physiology. Relatedly, while many genes necessary for HDR are essential for HAP1 survival (**Fig. 1a, Extended Data Fig. 1**), we cannot be certain it is *BRCA1*'s role in that pathway that makes it essential in HAP1. Nonetheless, the strong agreement we observe with assays for HDR and transcriptional activation (**Extended Data Fig. 9m,n**) suggests that at least in the regions assayed, relatively few variants impact specific functions of *BRCA1* but not others. Another limitation is that we are not currently able to interrogate every possible SNV in the targeted exons. Of note, most of the 3.5% of SNVs without SGE function scores were excluded by factors related to genome editing, rather than because of sampling (**Extended Data Fig. 6**). Lastly, as these experiments were designed to measure loss-of-function in a haploid cell line, we are unable to detect all types of functional effects (*e.g.* dominant negative variants).

SUPPLEMENTARY REFERENCES

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SUPPLEMENTARY TABLE GUIDE

Supplementary Table 1 | Saturation genome editing scores for 3,893 BRCA1 SNVs. Data were derived from experiments in HAP1-Lig4KO cells (except for columns labeled 'WT'). Scores are provided for all SNVs passing filters. SNV counts are from sequencing reads perfectly matching the reference sequence except for the SNV and either one or two 'HDR marker' variants. RNA scores are included for exonic SNVs assayed except for those in exon 18. SNV annotations from ClinVar and allele frequencies from the gnomAD, Bravo, and FLOSSIES databases are included (see Methods for full description). WT HAP1 SGE function scores are included for 3,407 SNVs passing the same filters (described in Extended Data Fig. 6) and present in the HAP1-Lig4KO data set as well. Abbreviations: 'reference' = reference nucleotide (genomic), 'alt' = alternative nucleotide (genomic), 'transcript_position' = position on reference transcript NM_007294.3, 'aa' = amino acid, 'aa_pos' = amino acid position, 'consequence' = mutational consequence in relation to the gene, 'function.score.mean' = mean function score across two biological replicates, 'func.class' = functional class assigned, 'p.nonfunctional' = mixture model point estimate of probability function score from non-functional distribution, 'experiment' = exon region assayed, 'neg./negative' = negative control, 'lib./library' = SNV library, 'd5' = day 5, 'd11' = day 11, 'r1/r2' = replicate 1/2, 'freq' = proportion of sequencing reads containing SNV, 'loess.fit' = LOESS regression value to model positional biases, 'aGVGD' = AlignGVGD, 'aGVGD.diff' = Grantham variation - Grantham deviation, 'aGVGD.class' = AlignGVGD classification, 'clinvar_simple' = ClinVar classification used for analysis, 'AF' = allele frequency, 'NA' = Not applicable.

Supplementary Table 2 | Analysis of SNVs with conflicting interpretations in ClinVar. The 122 SNVs assayed with SGE that had conflicting interpretations in ClinVar as of January 2, 2018 are shown here. Conflicts were assessed by reviewing interpretations from individual submitters. Columns for variant 'Name', 'Gene(s)', 'Condition(s)', 'Clinical significance', 'Review status', chromosomal coordinates, Variation ID, and Allele ID were downloaded from ClinVar. The 'BIC_influenced_conflict' column indicates whether each variant would be without conflicting interpretations if the BIC submission was removed, and likewise for the 'OMIM_influenced_conflict' column. The 'clinvar_without_BIC' column indicates what the interpretation would be without any BIC submissions ('P' = pathogenic, 'LP' = Likely pathogenic, 'B' = Benign, 'LB' = Likely benign, 'CI' = Conflicting interpretation, 'VUS' = Variant of uncertain significance). 'Conflicting_interpretations_provided' shows the various interpretations for each variant currently in ClinVar (as of 5/1/2018), and the 'best_guess' column indicates whether more individual interpretations provided favor 'pathogenic' or 'benign' status. SGE functional class assignments are shown for each variant ('FUNC' = functional, 'INT' = intermediate, and 'LOF' = non-functional), and average function scores across HAP1-Lig4KO replicates are provided for variants in the intermediate category. Additional sheets show the distributions for each 'best_guess' classification or 'clinvar_without_BIC' classification by SGE functional class. Sensitivity and specificity values of SGE function scores for predicting 'best_guess' classifications and 'clinvar_without_BIC' classifications are calculated using the cutoff determined by the ROC curve analysis in Fig. 3b.

Supplementary Table 3 | DNA sequences used in BRCA1 saturation genome editing experiments. Sequences are provided in the 5' to 3' orientation. Abbreviations used in primer naming are as follows: 'x' = exon; 'SGE' = sequence in which to introduce all possible SNVs for designing array-synthesized oligos (contains HDR marker substitutions); 'Array' = primer to amplify oligo pool from array (homologous to

non-mutagenized region present on each array oligo); 'LIN' = primer to perform inverse PCR to linearize homology arm constructs; 'gRNA_F' / 'gRNA_R' = oligos to anneal and clone into gRNA constructs; 'PU1(L/R)_Seq' = Custom sequencing primer for each amplicon; 'ext' = primer to amplify gDNA from cells, external to homology arms; 'SeqOut' = primer to amplify gDNA from cells, for use with 'ext'; 'HA' = primer used to amplify homology arms from HAP1 genomic DNA; 'pUC19' = primer with overhang providing homology for cloning HDR arms into pUC19; 'cDNA' = primer used for amplicon sequencing of BRCA1 cDNA; 'Ix' = Custom sample indexing primer with flow cell adapters for amplicon sequencing, compatible with PU1 sequencing primers. All other primers are annotated with notes.